# An Improved YOLO26 Algorithm for Small-Target Detection Based on Image Enhancement

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***Abstract*—To address the issues of high-frequency detail loss, insufficient spatial positional information, and inadequate multi-scale feature utilization in small-target detection under complex scenes, we propose a lightweight object detection method integrating wavelet feature enhancement and a dual-branch collaborative feature modeling mechanism that jointly captures frequency-domain details and global semantic context. First, a wavelet transform module is introduced at the feature extraction stage to decompose and reconstruct features from the frequency domain perspective, effectively enhancing the representation of high-frequency details such as edges and textures and compensating for the deficiencies of conventional convolution in fine-grained modeling. Second, a collaborative modeling mechanism between the wavelet enhancement module and the C2PSA structure is constructed; by fusing channel dependencies and spatial attention, efficient interaction between local details and global semantic information is achieved, improving the discriminability and stability of feature representation. Finally, to mitigate the weakening response of small targets in deep features, a dedicated small-target detection head is designed with a high-resolution feature branch to enhance the perception of tiny targets. Experimental results on the VisDrone dataset demonstrate that the proposed method effectively improves small-target detection accuracy and overall performance while maintaining model lightweightness and real-time capability, exhibiting promising application potential.**

***Keywords— VisDrone; wavelet transform, feature enhancement, collaborative modeling, high-resolution detection head***

## I. Introduction

In complex visual scenarios, small-target detection remains one of the key challenges limiting performance improvement in object detection. Particularly from aerial perspectives, targets typically exhibit characteristics of small scale, dense distribution, and strong background interference, making effective feature representation difficult. Although deep learning-based object detection methods have achieved significant progress in recent years, obvious bottlenecks remain in small-target scenarios. On the one hand, convolutional neural networks (CNNs) tend to attenuate high-frequency detail information during multi-layer downsampling, making it difficult to preserve target edge and structural features [2]. On the other hand, existing feature modeling approaches still exhibit deficiencies in spatial positional information representation and multi-scale information fusion, making it difficult for models to establish stable and reliable discriminative evidence for small targets. These issues are particularly pronounced in complex aerial benchmark datasets represented by VisDrone [1], becoming the main obstacle restricting further improvement of detection accuracy and robustness.

To this end, we improve the lightweight detection framework from two aspects: feature enhancement and structural collaborative modeling. First, we introduce a wavelet transform-based feature enhancement module to perform multi-subband decomposition and reconstruction on input features from the frequency domain perspective, explicitly enhancing high-frequency detail representation and compensating for the inherent deficiencies of conventional convolution in fine-grained feature modeling [3]. Second, we construct a collaborative modeling mechanism between the wavelet enhancement module and the C2PSA structure [4]; through the joint drive of channel dependency modeling and spatial attention, efficient interaction between local detail features and global semantic information is promoted, improving the discriminability and stability of feature representation. Finally, to address the weakening response of small targets in deep features, we design a dedicated small-target detection head and introduce a high-resolution feature branch to enhance the perception capability for tiny targets. Under the premise of maintaining model lightweightness and real-time performance, the proposed method achieves superior small-target detection performance in complex scenarios.

## II. Related Work

Mainstream object detection paradigms can be broadly divided into Transformer-based models represented by DETR [5] and single-stage convolutional frameworks represented by YOLO [6]. The former utilizes global self-attention to model


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long-range dependencies and capture rich semantic information, while the latter features a concise structure and efficient inference, making it more suitable for real-time applications. With continuous improvements in feature fusion and attention mechanisms, the YOLO series [7] has become a mainstream choice for practical deployment. However, in complex aerial datasets such as VisDrone, challenges including extremely small targets, dense distributions, and significant background interference still limit detection performance.

To address these issues, we improve the YOLO26 framework from both feature enhancement and structural design perspectives. A wavelet transform-based module is introduced to enhance high-frequency details during feature extraction, compensating for the limitations of conventional convolution in representing edge and texture information [8]. In addition, a collaborative mechanism combining the wavelet module and the C2PSA structure is designed to jointly model channel and spatial dependencies. Finally, a dedicated small-target detection head [9], [10] with a high-resolution branch is incorporated to improve the representation of tiny objects, achieving better detection performance while maintaining lightweight and real-time characteristics.

## III. Method

### A. Image Enhancement

We propose an image feature enhancement module based solely on wavelet transform, termed Wave, which performs frequency-domain decomposition on spatial features to explicitly extract low-frequency approximation information and multi-directional high-frequency detail information, and fuses them back into the original feature space to enhance the model's perception capability for multi-scale structural information. This module differs from any attention mechanism, aiming to verify the contribution of wavelet transform itself to feature representation through ablation experiments. The overall processing pipeline of the module is shown in Fig. 1.

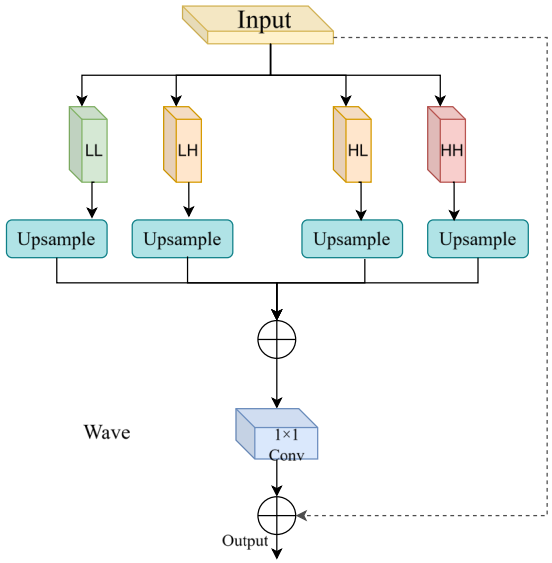



Figure 1 Overall pipeline of the proposed Wave module for frequency-domain feature enhancement

Given an input feature map $\mathbf{x} \in \mathbb{R}^{B\times C\times H\times W}$, it sequentially undergoes 2D discrete wavelet transform, sub-band upsampling, feature fusion, and residual connection, finally outputting an enhanced feature map with the same dimensions as the input.

We adopt a convolution-based differentiable 2D discrete wavelet transform (2D-DWT) using Haar wavelets as the basis functions. The low-pass and high-pass decomposition filters of the Haar wavelet are defined as in Equation (1):

$$\mathbf{h}_L = \frac{1}{\sqrt{2}}[1 \quad 1], \mathbf{h}_H = \frac{1}{\sqrt{2}}[1 \quad -1] \tag{1}$$

For the input feature map $\mathbf{x}$, the 2D wavelet transform constructs four 2D convolutional kernels through the outer product of the row and column filters, as in Equations (2) (3):

$$\mathbf{K}_{LL} = \mathbf{h}_L^{\top} \otimes \mathbf{h}_L, \mathbf{K}_{LH} = \mathbf{h}_L^{\top} \otimes \mathbf{h}_H \tag{2}$$

$$\mathbf{K}_{HL} = \mathbf{h}_H^{\top} \otimes \mathbf{h}_L, \mathbf{K}_{HH} = \mathbf{h}_H^{\top} \otimes \mathbf{h}_H \tag{3}$$

Where $\otimes$ denotes the vector outer product. The decomposition results of the four sub-bands are computed via channel-wise grouped convolution (stride = 2), as in Equation (4):

$$\mathbf{S}_{ij} = \mathbf{x} *_g \mathbf{K}_{ij}, ij \in \{LL, LH, HL, HH\} \tag{4}$$

Where $*_g$ denotes grouped convolution (groups = C), and the output sub-bands have spatial dimensions of $H/2 \times W/2$ with the channel number preserved as $C$.

The physical interpretations of the four sub-bands are as follows: $\mathbf{S}_{LL}$ is the low-frequency approximation sub-band, preserving the overall structure and smooth region information of the image; $\mathbf{S}_{LH}$ is the horizontal high-frequency sub-band, corresponding to horizontal edges in the image; $\mathbf{S}_{HL}$ is the vertical high-frequency sub-band, corresponding to vertical edges; $\mathbf{S}_{HH}$ is the diagonal high-frequency sub-band, corresponding to texture details and corner information.

Due to the stride of 2 in wavelet decomposition, the spatial resolution of each sub-band is reduced to half that of the original feature map. To integrate the multi-band information back to the original scale, each sub-band is upsampled via bilinear interpolation, as in Equation (5):

$$\hat{\mathbf{S}}_{ij} = \text{Upsample}(\mathbf{S}_{ij}, \text{size} = (H, W), \text{mode=bilinear}) \tag{5}$$

The four upsampled sub-bands are concatenated along the channel dimension to obtain the multi-frequency fused feature, as in Equation (6):

$$\mathbf{F}_{cat} = \text{Concat}\left([\hat{\mathbf{S}}_{LL}, \hat{\mathbf{S}}_{LH}, \hat{\mathbf{S}}_{HL}, \hat{\mathbf{S}}_{HH}], \text{dim} = 1\right) \in \mathbb{R}^{B\times 4C\times H\times W} \tag{6}$$

To compress the concatenated feature with quadruple channels back to the original channel number, a $1 \times 1$ convolution is adopted for linear channel fusion, followed by a Batch Normalization (BN) layer to stabilize the training process, as in Equation (7):


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$$\mathbf{F}_{wav} = \mathrm{BN}(\mathrm{Conv}_{1\times1}(\mathbf{F}_{cat})) \in \mathbb{R}^{B\times C\times H\times W} \quad (7)$$

Where the convolutional kernel parameters $\mathbf{W} \in \mathbb{R}^{C\times 4C\times 1\times 1}$ of $\mathrm{Conv}_{1\times1}$ learn the weighted fusion strategy for multi-band features through end-to-end training, with no bias applied (bias=False).

Finally, the module output $\mathbf{y}$ is obtained via a residual connection. The formulation is conditional on channel consistency: when the input and output channel dimensions are equal (C_in = C_out), the residual addition is applied directly, as in Equation (8):

$$\mathbf{y} = \mathbf{x} + \mathbf{F}_{wav} \quad (8)$$

When the channel dimensions differ (C_in ≠ C_out), a 1×1 convolution is inserted on the residual path to align channels before addition, as in Equation (9):

$$\mathbf{y} = \mathbf{Conv1\times1}(\mathbf{x}) + \mathbf{F}_{wav} \quad (9)$$

Equations (8) and (9) thus represent the two conditional cases of the same residual connection design; Equation (9) reduces to Equation (8) when no channel projection is needed.

To ensure gradient computability of the wavelet transform during backpropagation, the convolution-based DWT implementation (WaveletTransform2D) directly leverages the PyTorch autograd mechanism, inherently satisfying differentiability. For the reference implementation using PyWavelets (WaveletTransform2D_PyWT), since its underlying operations are NumPy-based, a custom gradient function WaveletGrad is introduced. During forward propagation, sub-band coefficients are passed through, while during backpropagation, gradients are approximately backpropagated via bilinear upsampling, as formulated in Equation (10):

$$\frac{\partial \mathcal{L}}{\partial \mathbf{x}} \approx \mathit{Upsample}\left(\frac{\partial \mathcal{L}}{\partial \mathbf{s}_{ij}}, \mathrm{size} = (H, W)\right) \quad (10)$$

This approximate design ensures reasonable gradient propagation during training while avoiding the implementation complexity of exact differentiation for the inverse transform of PyWavelets.

### B. *Collaborative Architecture*

To further exploit the complementary potential between wavelet-enhanced features and semantic features, a collaborative modeling mechanism between the wavelet enhancement module and the C2PSA structure, termed the C2PSA-CW module, is constructed based on the Wave module. The design motivation is as follows: although pure wavelet transform can effectively strengthen high-frequency detail information, its essence is linear frequency-domain decomposition, lacking the capability to model channel-wise dependencies and global spatial context; whereas the C2PSA structure can capture global semantic information through position-sensitive attention mechanisms, but its perception capability for fine edges and texture details is relatively limited. The two form a natural complementarity at the feature modeling level, and through collaborative fusion, efficient interaction between local details and global semantics can be achieved [11]. The overall structure of the C2PSA-CW module is shown in Fig. 2.

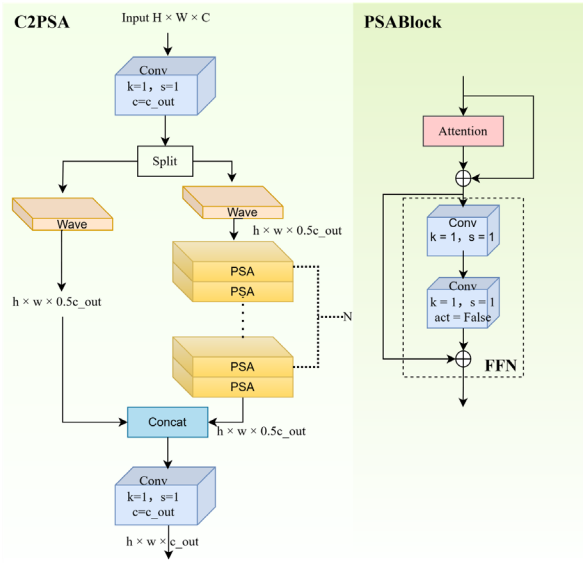



Figure 2. Architecture of the proposed Cw2PSA collaborative module

Given an input feature map $\mathbf{x} \in \mathbb{R}^{B\times C\times H\times W}$, it first undergoes $1\times1$ convolution for channel compression, mapping the channel number from $C$ to $C_{out}$, and is then split into two branches along the channel dimension, each occupying $0.5C_{out}$ channels.

One branch enters the wavelet enhancement branch (Wave Branch), where the split features are fed into the Wave module for frequency-domain decomposition and reconstruction, extracting multi-directional high-frequency detail information and completing residual fusion, as in Equation (11):

$$\mathbf{F}_{wave} = \mathrm{Wave}(\mathbf{x}_{split1}) \in \mathbb{R}^{B\times 0.5C_{out}\times H\times W} \quad (11)$$

The other branch enters the PSA attention branch (PSA Branch), undergoing global semantic modeling via $N$ stacked PSABlocks. Each PSABlock comprises a multi-head self-attention module (Attention) and a feed-forward network (FFN) [12], and fuses attention weights through two parallel $1\times1$ convolutional paths to achieve joint modeling of channel dependencies and spatial relationships, as in Equation (12):

$$\mathbf{F}_{psa} = \mathrm{PSABlock}^{N}(\mathbf{x}_{split2}) \in \mathbb{R}^{B\times 0.5C_{out}\times H\times W} \quad (12)$$

The outputs of the two branches are concatenated along the channel dimension and passed through $1\times1$ convolution to complete the final channel fusion and output mapping, as in Equation (13):

$$\mathbf{y} = \mathrm{Conv}_{1\times1}\big(\mathit{Concat}([\mathbf{F}_{wave}, \mathbf{F}_{psa}], \mathrm{dim} = 1)\big) - \in \mathbb{R}^{B\times C_{out}\times H\times W} \quad (13)$$

Through the above collaborative mechanism, the edge and texture detail information provided by the wavelet branch and the global semantic context information provided by the PSA


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branch achieve complementary fusion within the same feature space, enabling the final output features to simultaneously possess fine-grained local perception and cross-regional semantic association capabilities. This design effectively improves the discriminability and stability of feature representation without significantly increasing computational overhead, particularly facilitating the establishment of more reliable detection evidence in aerial scenes with complex backgrounds and tiny targets.

### C. Small-Target Detection Architecture

In complex aerial scenes, small targets occupy extremely limited pixels, and their features are easily weakened during multi-level downsampling. Based on the YOLO26 framework [13], the detection architecture is improved from three aspects: backbone enhancement, multi-scale feature fusion, and small-target detection head design.

For the backbone, the original C2PSA module is replaced with the Cw2PSA module, which integrates wavelet-based frequency-domain enhancement to preserve high-frequency details while maintaining semantic representation. Meanwhile, a Wave module is embedded into each stage as a bypass unit to compensate for information loss caused by downsampling. In the neck, a feature pyramid structure with top-down upsampling and lateral connections is adopted to achieve effective multi-scale feature fusion [2]. For detection, an additional small-target head (stride 4) is introduced alongside the original three heads, utilizing high-resolution features to improve the response to tiny objects. The four detection scales (160×160, 80×80, 40×40, and 20×20) jointly cover targets of different sizes, and the overall structure is illustrated in Fig. 3.

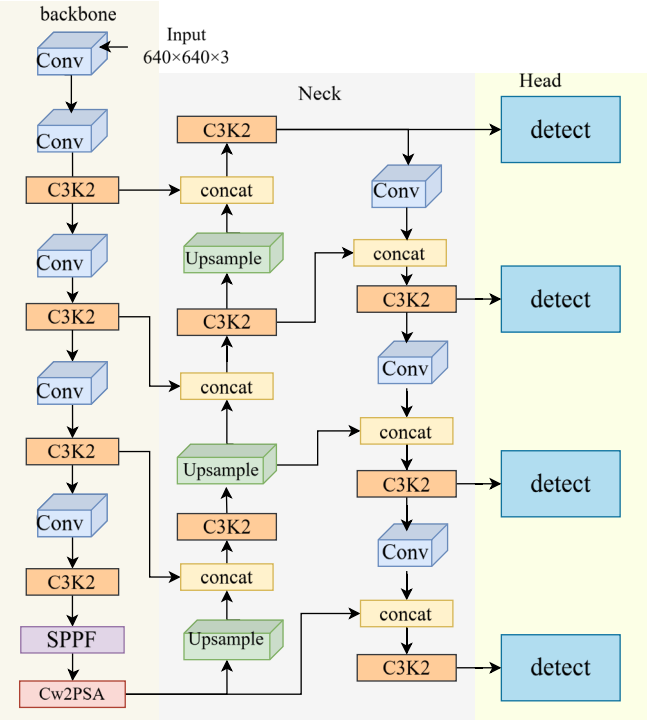



Figure 3 Overall architecture of the proposed detection network

In summary, the proposed architecture systematically strengthens the perception and detection capability for small targets in complex aerial scenes through the organic combination of frequency-domain feature enhancement, collaborative attention modeling, and high-resolution detection heads, while maintaining model lightweightness and inference efficiency.

## IV. Experiments and Analysis

Experiments were conducted on a Windows 11 system equipped with an Intel Core i5-14600KF processor and an NVIDIA GeForce RTX 4070 SUPER 12 GB GPU, using PyTorch 2.5.1 + TorchVision 0.20.1 with CUDA 12.1.

### A. Datasets and Evaluation Metrics

The VisDrone-DET2019 dataset [1] was adopted as the data source for training and evaluation. VisDrone is a large-scale benchmark collected by UAVs across 14 different cities in China, comprising 6,471 training images, 548 validation images, and 1,580 test images, with annotations covering 10 object categories (pedestrian, person, car, van, bus, truck, motor, bicycle, awning-tricycle, and tricycle). The dataset is publicly available at： https://github.com/VisDrone/VisDrone-Dataset. During data preprocessing and training, input images were uniformly resized to 640 × 640 pixels, and sample diversity was enhanced through Mosaic augmentation (probability 1.0), mixup scaling (scale: 0.5), and horizontal flipping (flip: 0.5). Model training employed the stochastic gradient descent (SGD) optimizer with momentum, with an initial learning rate of 0.01. The batch size was set to 8 under available GPU memory conditions, and the model was trained for 200 epochs to ensure full convergence.

To systematically evaluate model performance on small-target detection tasks, Precision, Recall, mean Average Precision at an IoU threshold of 0.5 (mAP@50), and mean Average Precision over the 0.5–0.95 interval (mAP@50–95) were selected as the primary evaluation metrics. Model parameters (Parameters) and floating-point operations (FLOPs) were also incorporated for comprehensive analysis of computational complexity and efficiency. Intersection over Union (IoU) measures the overlap between predicted and ground-truth bounding boxes; a prediction is counted as a True Positive (TP) when IoU exceeds the set threshold, otherwise as a False Positive (FP). Based on this, the evaluation metrics are further calculated, as formulated in Equation (14)-(18).

$$IoU = \frac{B_p \cap B_{gt}}{B_p \cup B_{gt}} \tag{14}$$

$$Precision = \frac{TP}{TP+FP} \tag{15}$$

$$Recall = \frac{TP}{TP+FN} \tag{16}$$

$$mAP = \frac{1}{C}\sum_{i=1}^{C} AP_i \tag{17}$$

$$mAP@50\text{--}95 = \frac{1}{10}\sum_{k=0}^{9} mAP_{0.5+0.05k} \tag{18}$$


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### B. Comparative Experiments

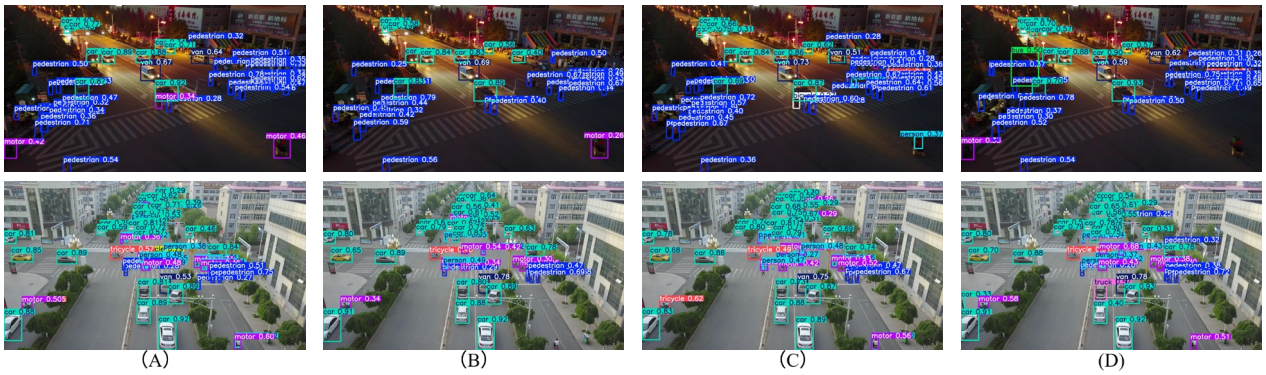
To verify the detection efficacy of the proposed model, multiple lightweight YOLO series models and other improved schemes were selected for comparative experiments. Detection accuracy and computational complexity were evaluated under unified configurations, with results shown in Table I and detection visualizations shown in Fig. 4.

As shown in the table, the proposed method achieves a significant leap in detection performance while ensuring model lightweightness. Compared with mainstream YOLO series baseline models (from YOLOv8n to YOLOv12n) and other comparison schemes such as BGF and OSD, the proposed method reaches the optimal level across four core evaluation metrics: mAP50-95, mAP50, Precision, and Recall. Notably, although the model parameters (2.7M) and computational FLOPs (6.6G) are not the absolute lowest overall, obvious computational cost reduction has been achieved compared with models such as YOLOv8n. Furthermore, compared with YOLO26n or OSD models pursuing extreme lightweightness, the proposed method trades only a slight increase in parameters and computational cost for highly competitive accuracy gains (e.g., mAP50-95 improves by approximately 10.4% over YOLO26n). This represents an excellent balance between feature representation capability and model complexity. As shown in the table, the proposed method achieves a significant leap in detection performance while ensuring model

### C. Ablation Study

To quantitatively verify the independent contributions of the proposed improvement modules to final detection performance, ablation experiments were conducted on the VisDrone dataset with YOLO26n as the baseline model, progressively introducing each core component and recording performance changes, with results shown in Table II.

Four configurations were established, progressively introducing the wavelet enhancement module (Wave), the collaborative modeling module (Cw2PSA), and the full model (Full Model) on the baseline. Results show that the introduction of each module brings continuous performance improvement: after adding the wavelet enhancement module alone (Configuration B), the mAP50-95 improves from 0.183 to 0.191 (+4.4%) and mAP50 from 0.325 to 0.328, providing intermediate quantitative evidence that the frequency-domain decomposition itself contributes positively to feature representation, independent of the attention mechanism introduced in Configuration C; upon further introducing the collaborative modeling module, accuracy continues to improve, indicating significant synergistic gains between wavelet enhancement and attention mechanisms; the full model achieves optimal results across all metrics, trading reasonable parameters and computational cost for the best detection performance, fully validating the necessity of each module design and the effectiveness of their combination.



(A) (B) (C) (D)

Figure 4 Visual comparison of detection results, where (A) denotes Ours, and (B), (C), and (D) represent YOLO26, YOLOv12, and YOLOv10, respectively.

TABLE I COMPARATIVE EXPERIMENTS

| Model Name | mAP50-95 | mAP50 | Precision | Recall | Parameters(M) | FLOPs（G） |
|---|---|---|---|---|---|---|
| ***YOLOv8n[20]*** | 0.192 | 0.328 | 0.435 | 0.328 | 3.2 | 8.8 |
| ***YOLOv9t[14]*** | 0.194 | 0.333 | 0.433 | 0.326 | 2.0 | 7.8 |
| ***YOLOv10n[15]*** | 0.196 | 0.327 | 0.431 | 0.330 | 2.3 | 8.4 |
| ***YOLOv11n[16]*** | 0.185 | 0.329 | 0.418 | 0.328 | 2.6 | 6.4 |
| ***YOLOv12n[17]*** | 0.185 | 0.327 | 0.419 | 0.333 | 2.6 | 6.5 |
| ***YOLO26n[13]*** | 0.183 | 0.325 | 0.426 | 0.325 | 2.5 | **5.8** |
| ***BGF[18]*** | 0.194 | 0.320 | 0.416 | 0.320 | 2.0 | 8.6 |
| ***OSD[19]*** | 0.192 | 0.334 | 0.438 | 0.328 | **1.6** | 7.9 |
| ***Ours*** | **0.202** | **0.339** | **0.448** | **0.345** | 2.7 | 6.6 |

TABLE II ABLATION STUDY

| Model | Wave | Cw2PSA | Full Model | mAP50-95 | mAP50 | Precision | Recall | Params(M) | FLOPs(G) |
|---|---|---|---|---|---|---|---|---|---|
| A | | | | 0.183 | 0.325 | 0.426 | 0.325 | 2.5 | **5.8** |
| B | ✓ | | | 0.191 | 0.328 | 0.431 | 0.330 | 2.5 | 5.9 |
| C | ✓ | ✓ | | 0.197 | 0.331 | 0.439 | 0.338 | 2.6 | 5.9 |
| D | ✓ | ✓ | ✓ | **0.202** | **0.339** | **0.448** | **0.345** | 2.7 | 6.6 |



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## V. Conclusion and Future Work

To address the core issues of high-frequency detail loss, insufficient spatial modeling, and inadequate multi-scale feature utilization in small-target detection under complex aerial scenes, we propose an improved detection framework integrating wavelet feature enhancement and collaborative attention modeling. By introducing a differentiable wavelet transform module, constructing a frequency-domain-attention collaborative modeling mechanism, and designing a dedicated small-target detection head, the proposed method effectively improves detection accuracy and feature representation capability for small targets in complex scenes while maintaining model lightweightness and real-time inference, validating the effectiveness and synergistic gains of each module design.

In future work, on the one hand, the fixed Haar wavelet basis can be explored to be replaced with learnable wavelet bases, enabling the frequency-domain decomposition strategy to adaptively optimize for specific tasks, and multi-level wavelet decomposition can be introduced to more fully exploit structural features at different scales; on the other hand, the generalization capability of the current method needs further validation on more diverse datasets and deployment scenarios (e.g., satellite remote sensing, embedded edge platforms) to clarify its applicable boundaries and provide specific guidance for further improvements.

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